Experiment 4-B

Vehicle Service Request Form Using Advanced Fields

Aim

To extend the Vehicle Service Request Form in Zoho Creator by incorporating all seventeen Advanced Fields of the Form Builder, so that the form supports service-history review, cost computation, vehicle model verification, media handling and automated service-ID / QR generation.

Procedure

1. Owner completes the existing basic-field sections (personal, contact, vehicle and issue details).
2. Front office verifies the Vehicle Model using the Lookup field, which auto-fills the standard service package and estimated duration.
3. The vehicle's Previous Service History is reviewed or added as rows in the SubForm for technician reference.
4. Owner uploads the insurance copy, damage photo and, optionally, an issue video and a voice note of any unusual sound.
5. Owner indicates whether pickup/drop is required; the Estimated Service Cost, Service Discount and computed Net Service Cost Payable are displayed.
6. Owner reviews the Service Center Guidelines block and provides a Digital Signature authorizing the estimated work.
7. On Submit, Zoho Creator generates the Service Request ID, stamps the Submitted On date-time and generates the Service Tracking QR Code.
8. The service center reviews Service_Request_Report, verifies uploads and history, and uses the QR code to track the vehicle through the repair pipeline.

Advanced Fields Used

Advanced Field	Field Name	Purpose / Sample
Decision Box	Requires_Pickup_Drop_Service	Yes/No: does the owner need pickup and drop
Rich Text	Service_Center_Guidelines_Display	Formatted guidelines shown inside the form
URL	Insurance_Policy_Portal_URL	Link to online motor insurance policy
Image	Vehicle_Damage_Photo	Photo of damaged / faulty area (JPG, PNG)
Percent	Service_Discount_Percent	Loyalty or offer discount, e.g. 10%
Currency	Estimated_Service_Cost	Estimated cost, e.g. ₹4,200.00
Decimal	Vehicle_Age_Years	Vehicle age in years, e.g. 3.5
Date-Time	Service_Request_Submitted_On	Auto-captured submission timestamp
SubForm	Previous_Service_History	Rows of date, service type and cost
Lookup	Vehicle_Model_Lookup	Auto-fills package and duration from Vehicle Models form
Signature	Owner_Digital_Signature	Owner's drawn authorization
File Upload	Insurance_Copy_Upload	Scanned insurance copy (PDF, JPG, PNG)
Audio	Engine_Sound_Voice_Note	Optional clip of unusual engine or brake sound

Advanced Field	Field Name	Purpose / Sample
Video	Vehicle_Issue_Video	Optional short video of the reported issue
Auto Number	Service_Request_ID	SVC-REQ-2026-000691 style unique ID
Formula	Net_Service_Cost_Payable	Estimated cost minus (cost × discount %)
QR / Barcode	Service_Tracking_QR_Code	QR encoding the Service Request ID

Extended Form Layout

- **Section E, Vehicle Verification & Evidence:** Vehicle Model Lookup, Vehicle Damage Photo, Insurance Copy Upload, Vehicle Age in Years
- **Section F, Service History & Media Details:** Previous Service History (SubForm), Insurance Policy Portal URL, Vehicle Issue Video, Engine Sound Voice Note
- **Section G, Cost & Pickup Preference:** Requires Pickup/Drop Service, Estimated Service Cost, Service Discount Percent, Net Service Cost Payable
- **Section H, Authorization & System-Generated Values:** Guidelines Display, Owner Digital Signature, Submitted On, Service Request ID, Service Tracking QR Code

Validation & Configuration Rules

- Lookup: connected to a master Vehicle Models form; staff search and select instead of typing free text.
- SubForm: minimum 0 and maximum 10 rows per vehicle.
- File Upload: PDF, JPG and PNG only, maximum 5 MB.
- Image: JPG and PNG only, maximum 2 MB.
- Video: maximum duration 45 seconds and maximum size 50 MB.
- Percent: value range 0 to 25. Currency: non-negative values only, formatted in INR (₹).
- Formula: read-only, recalculated whenever Estimated Service Cost or Discount Percent changes.
- Auto Number: prefix **SVC-REQ-**, current year, and a 6-digit auto-incrementing sequence.
- QR/Barcode: generated automatically right after the Service Request ID is assigned.
- Signature and Decision Box are mandatory; the form cannot be submitted without them.

Output Screenshots

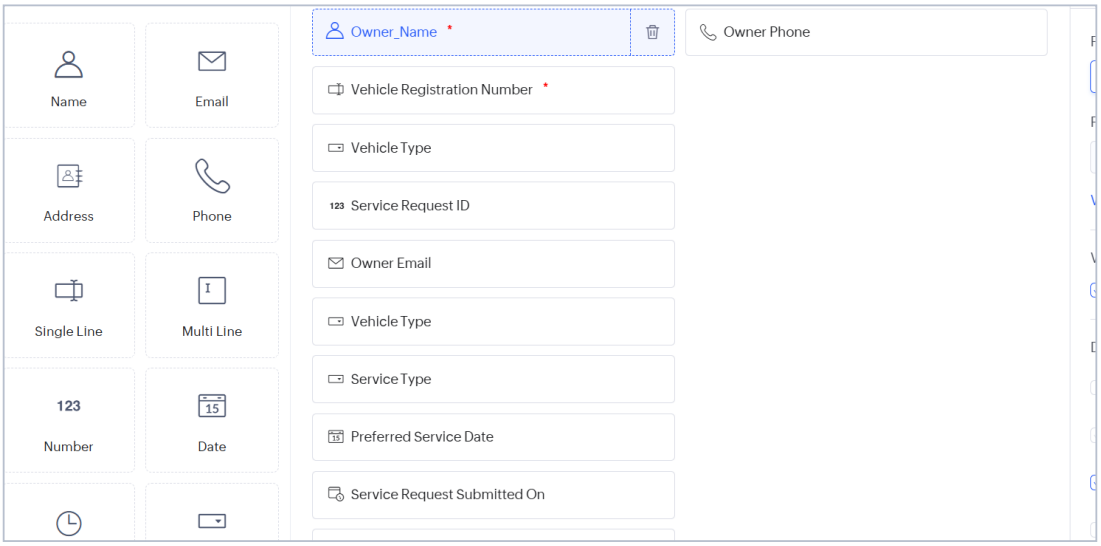


Figure 1: Form Builder: Service Request ID (Auto Number) and Service Request Submitted On (Date-Time) fields on the form.

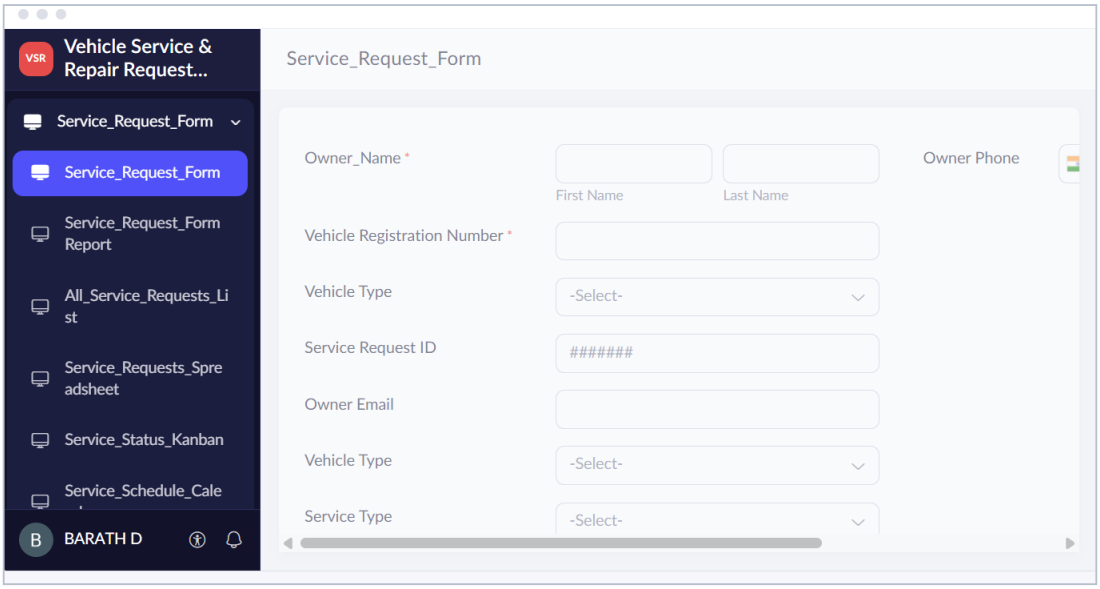


Figure 2: Live form: Service Request ID auto-number field (shown as ####### until submission).

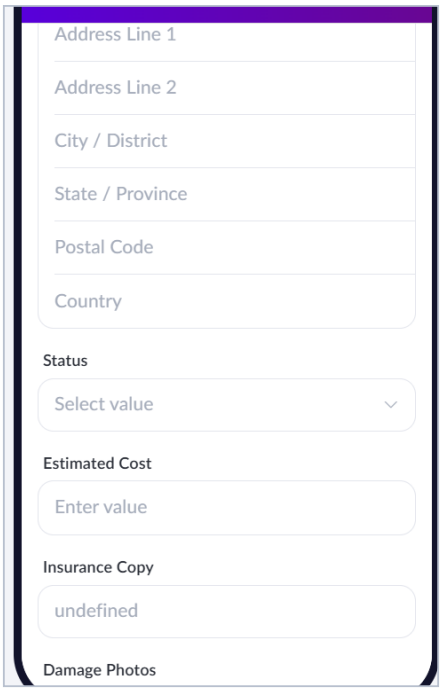


Figure 3: Mobile preview: Address, Status, Estimated Cost, Insurance Copy and Damage Photos fields.

Sample Record 1

Field	Value
Service_Request_ID (Auto Number)	SVC-REQ-2026-000691
Vehicle_Model_Lookup	Hyundai i20, Standard Service Package, 3 hours
Requires_Pickup_Drop_Service	Yes
Estimated_Service_Cost / Discount	₹4,200.00 / 10%
Net_Service_Cost_Payable (Formula)	₹3,780.00
Insurance_Copy_Upload	insurance_copy.pdf (within 5 MB)
Service_Tracking_QR_Code	QR encoding SVC-REQ-2026-000691

Sample Record 2

Field	Value
Service_Request_ID (Auto Number)	SVC-REQ-2026-000692
Vehicle_Model_Lookup	Maruti Swift, Standard Service Package, 2.5 hours
Requires_Pickup_Drop_Service	No
Estimated_Service_Cost / Discount	₹6,500.00 / 15%
Net_Service_Cost_Payable (Formula)	₹5,525.00
Service_Tracking_QR_Code	QR encoding SVC-REQ-2026-000692

Previous Service History (SubForm Rows)

Row	Service Date	Type of Service	Cost
1	12-Jan-2026	Oil Change	₹1,800
2	05-Apr-2026	Brake Pad Replacement	₹2,500

Formula Check: Net = Estimated − (Estimated × Discount %)

Estimated Cost	Discount %	Net Service Cost Payable
₹4,200	10%	₹3,780
₹6,500	15%	₹5,525
₹3,000	0%	₹3,000
₹8,000	25%	₹6,000

Validation Test Cases

Field	Test Input	Expected Result
Percent	30	Rejected; allowed range is 0 to 25
Currency	-500	Rejected; non-negative only
Image	damage.gif or file above 2 MB	Rejected; JPG and PNG up to 2 MB
File Upload	insurance.docx	Rejected; PDF, JPG, PNG up to 5 MB
Video	Clip longer than 45 seconds	Rejected; maximum 45 seconds / 50 MB
SubForm	11th row	Not allowed; maximum 10 rows
Signature / Decision Box	Left empty	Form cannot be submitted

Result

The Service_Request_Form was extended with all seventeen Advanced Fields, including lookup verification, service-history SubForm, computed net cost, auto-generated service ID and QR tracking code, and the validations were verified in Zoho Creator.